

1.2 Algebra & Functions

Laws of Indices

Indices (powers) follow a set of rules that allow us to simplify expressions involving powers.

Rule	Law	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$x^3 \times x^4 = x^7$
Division	$a^m \div a^n = a^{m-n}$	$x^5 \div x^2 = x^3$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
Zero index	$a^0 = 1$	$5^0 = 1$
Negative index	$a^{-n} = \frac{1}{a^n}$	$x^{-2} = \frac{1}{x^2}$
Fractional index	$a^{1/n} = \sqrt[n]{a}$	$8^{1/3} = 2$
General fractional	$a^{m/n} = (\sqrt[n]{a})^m$	$27^{2/3} = (\sqrt[3]{27})^2 = 9$

Example: Simplify $\frac{3x^{5/2} \times 2x^{-1}}{x^{1/2}}$

Numerator: $3x^{5/2} \times 2x^{-1} = 6x^{5/2-1} = 6x^{3/2}$

Divide by $x^{1/2}$: $6x^{3/2-1/2} = 6x$

Surds

A **surd** is an irrational root that cannot be simplified to a rational number, e.g. $\sqrt{2}$, $\sqrt{5}$, $\sqrt[3]{7}$.

Rule	Formula	Example
Multiplication	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$
Division	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$	$\frac{\sqrt{20}}{\sqrt{5}} = \sqrt{4} = 2$
Simplification	$\sqrt{a^2b} = a\sqrt{b}$	$\sqrt{50} = 5\sqrt{2}$
Rationalise (simple)	$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
Rationalise (compound)	Multiply by conjugate $\frac{a \mp \sqrt{b}}{a \mp \sqrt{b}}$	$\frac{1}{3 + \sqrt{2}} = \frac{3 - \sqrt{2}}{7}$

Example: Rationalise $\frac{5}{2 + \sqrt{3}}$

Multiply top and bottom by the conjugate $2 - \sqrt{3}$:

$$\frac{5}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{5(2 - \sqrt{3})}{4 - 3} = \frac{10 - 5\sqrt{3}}{1} = 10 - 5\sqrt{3}$$

Quadratics

A **quadratic expression** has the form $ax^2 + bx + c$ where $a \neq 0$. The graph is a parabola.

The Quadratic Formula

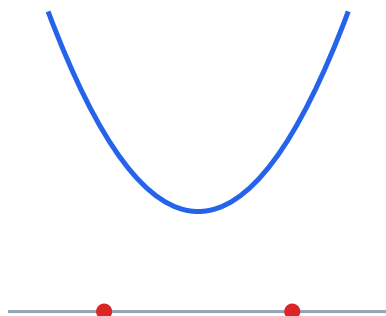
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Discriminant

The **discriminant** $\Delta = b^2 - 4ac$ determines the nature of the roots of a quadratic equation.

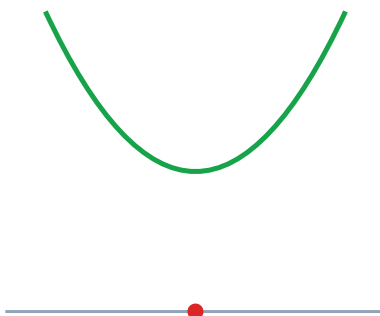
Discriminant	Nature of Roots	Graph
$b^2 - 4ac > 0$	Two distinct real roots	Parabola crosses x-axis twice
$b^2 - 4ac = 0$	One repeated real root	Parabola touches x-axis once
$b^2 - 4ac < 0$	No real roots (complex)	Parabola does not cross x-axis

Two distinct roots



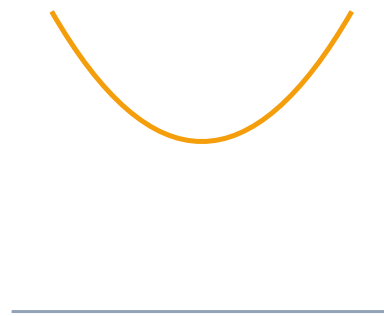
$$b^2 - 4ac > 0$$

One repeated root



$$b^2 - 4ac = 0$$

No real roots



$$b^2 - 4ac < 0$$

Completing the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c$$

Example: Write $2x^2 + 12x + 7$ in completed square form.

Factor out 2 from the x terms: $2(x^2 + 6x) + 7$

Complete the square inside: $2((x + 3)^2 - 9) + 7$

Expand: $2(x + 3)^2 - 18 + 7 = 2(x + 3)^2 - 11$

Vertex is at $(-3, -11)$. Minimum value is -11 .

Exam Tip: Completing the square tells you the vertex (turning point) of the parabola. If the question asks for the minimum/maximum value, this is the method to use. The vertex is

$$\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right).$$

Simultaneous Equations

Linear-Linear

Use **elimination** (add/subtract equations) or **substitution**.

Linear-Quadratic

Substitute the linear equation into the quadratic and solve the resulting quadratic.

Example: Solve $y = x + 3$ and $x^2 + y^2 = 29$.

Substitute: $x^2 + (x + 3)^2 = 29$

$$x^2 + x^2 + 6x + 9 = 29$$

$$2x^2 + 6x - 20 = 0 \Rightarrow x^2 + 3x - 10 = 0$$

$$(x + 5)(x - 2) = 0 \Rightarrow x = -5 \text{ or } x = 2$$

When $x = -5$: $y = -2$. When $x = 2$: $y = 5$.

Solutions: $(-5, -2)$ and $(2, 5)$.

Inequalities

Linear Inequalities

Solve like equations. **Reverse the inequality sign when multiplying/dividing by a negative number.**

Quadratic Inequalities

1. Solve the corresponding equation $ax^2 + bx + c = 0$ to find critical values.
2. Sketch the parabola.
3. Read off the solution from the graph.

Example: Solve $x^2 - 5x + 6 < 0$.

Factorise: $(x - 2)(x - 3) = 0$, so roots at $x = 2$ and $x = 3$.

The parabola is U-shaped (positive x^2 coefficient). It is below the x-axis between the roots.

Solution: $2 < x < 3$.

Exam Tip: Always sketch the quadratic to decide the correct region. For > 0 with a U-shaped parabola, the answer is two separate regions. For < 0 , the answer is between the roots.

Polynomials

Factor Theorem

If $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$.

Conversely, if $(x - a)$ is a factor of $f(x)$, then $f(a) = 0$.

Example: Show that $(x - 2)$ is a factor of $f(x) = x^3 - 6x^2 + 11x - 6$.

$f(2) = 8 - 24 + 22 - 6 = 0$. Since $f(2) = 0$, $(x - 2)$ is a factor. Then:

$$x^3 - 6x^2 + 11x - 6 = (x - 2)(x^2 - 4x + 3) = (x - 1)(x - 2)(x - 3)$$

Algebraic Division

Use polynomial long division or compare coefficients to divide a polynomial by a linear or quadratic factor.

Example: Divide $2x^3 + 3x^2 - 8x + 3$ by $(x + 3)$.

Using long division or inspection:

$$2x^3 + 3x^2 - 8x + 3 = (x + 3)(2x^2 - 3x + 1) = (x + 3)(2x - 1)(x - 1)$$

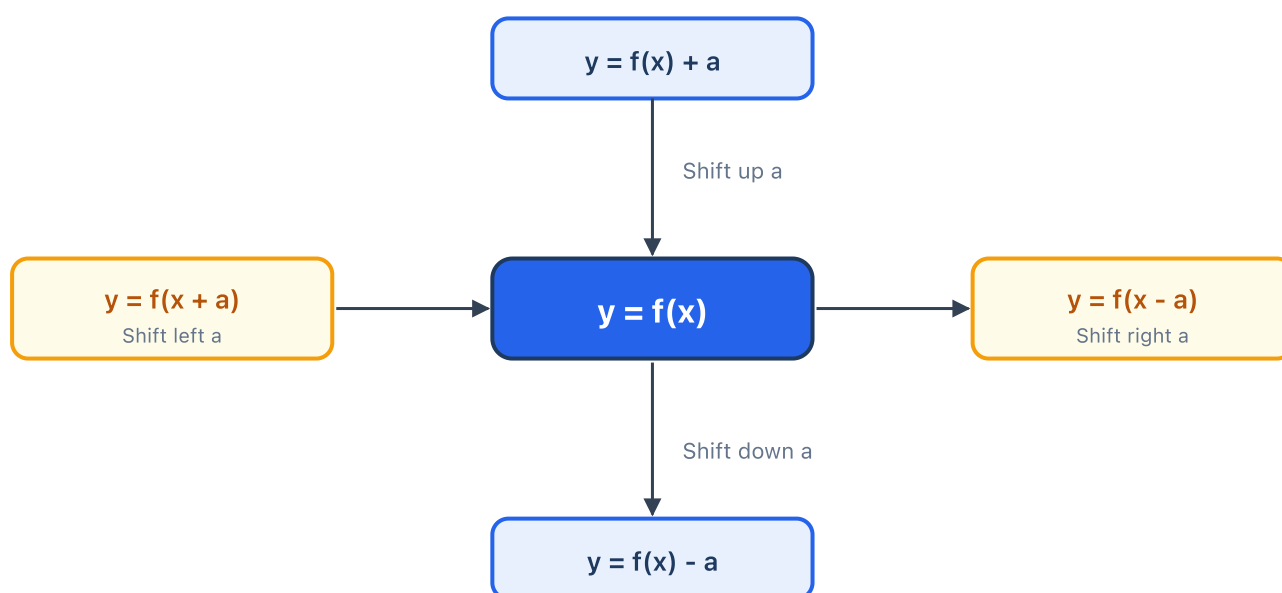
Graphs of Functions

You need to recognise and sketch the following key graphs:

- Linear: $y = mx + c$
- Quadratic: $y = ax^2 + bx + c$
- Cubic: $y = ax^3 + \dots$
- Reciprocal: $y = \frac{a}{x}$ and $y = \frac{a}{x^2}$
- Exponential: $y = a^x$
- Trigonometric: $y = \sin x, \cos x, \tan x$

Graph Transformations

Transformation	Notation	Effect
Translation up by a	$y = f(x) + a$	Shift graph up a units
Translation down by a	$y = f(x) - a$	Shift graph down a units
Translation right by a	$y = f(x - a)$	Shift graph right a units
Translation left by a	$y = f(x + a)$	Shift graph left a units
Vertical stretch, scale factor a	$y = af(x)$	Stretch parallel to y-axis by factor a
Horizontal stretch, scale factor $\frac{1}{a}$	$y = f(ax)$	Stretch parallel to x-axis by factor $\frac{1}{a}$
Reflection in x-axis	$y = -f(x)$	Flip graph over x-axis
Reflection in y-axis	$y = f(-x)$	Flip graph over y-axis



Exam Tip: Remember the counter-intuitive rule: changes *inside* the bracket (to x) do the *opposite* of what you expect. $f(x - 3)$ shifts **right** by 3, and $f(2x)$ squashes the graph **horizontally** by factor $\frac{1}{2}$.

Partial Fractions

Partial fractions decompose a single fraction into a sum of simpler fractions. This is the reverse of adding fractions together.

Types

Denominator Type	Partial Fraction Form
Linear factors: $\frac{f(x)}{(ax+b)(cx+d)}$	$\frac{A}{ax+b} + \frac{B}{cx+d}$
Repeated factor: $\frac{f(x)}{(ax+b)^2(cx+d)}$	$\frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{cx+d}$
Improper (degree num $\geq$ denom)	Divide first, then decompose the remainder

Example: Express $\frac{5x+1}{(x-1)(x+2)}$ **in partial fractions.**

$$\text{Let } \frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$$\text{Multiply through: } 5x+1 = A(x+2) + B(x-1)$$

$$\text{Set } x = 1: 6 = 3A \Rightarrow A = 2$$

$$\text{Set } x = -2: -9 = -3B \Rightarrow B = 3$$

$$\frac{5x+1}{(x-1)(x+2)} = \frac{2}{x-1} + \frac{3}{x+2}$$

Modulus Function

The **modulus** (absolute value) of x , written $|x|$, gives the magnitude of x without regard to sign.

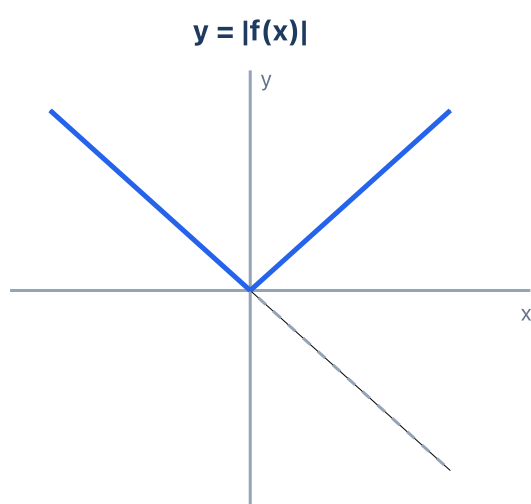
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Graph of $y = |f(x)|$

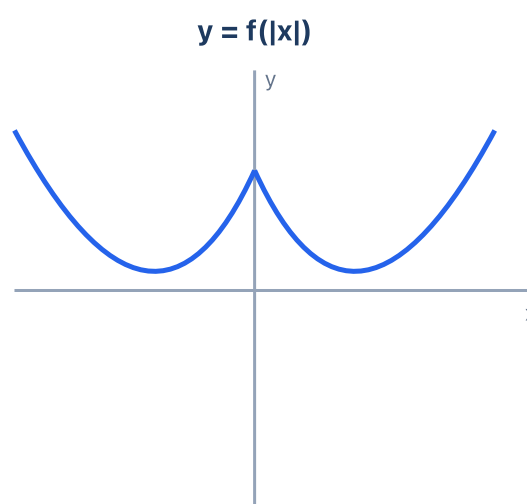
Reflect any part of $y = f(x)$ that is below the x-axis in the x-axis (make negative y-values positive).

Graph of $y = f(|x|)$

Keep the graph for $x \geq 0$ and reflect it in the y-axis (the graph becomes symmetric about the y-axis).



Reflect below x-axis upward



Symmetric about y-axis

Solving Modulus Equations

Example: Solve $|2x - 3| = 5$.

Case 1: $2x - 3 = 5 \Rightarrow 2x = 8 \Rightarrow x = 4$

Case 2: $2x - 3 = -5 \Rightarrow 2x = -2 \Rightarrow x = -1$

Solutions: $x = 4$ or $x = -1$.

Example: Solve $|x + 1| = |2x - 3|$.

Case 1: $x + 1 = 2x - 3 \Rightarrow 4 = x \Rightarrow x = 4$

Case 2: $x + 1 = -(2x - 3) \Rightarrow x + 1 = -2x + 3 \Rightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$

Solutions: $x = 4$ or $x = \frac{2}{3}$.

Exam Tip: When solving $|f(x)| = |g(x)|$, you can also square both sides: $f(x)^2 = g(x)^2$, which avoids case work. For modulus inequalities like $|f(x)| < a$, rewrite as $-a < f(x) < a$.

Functions: Domain and Range

- The **domain** of a function is the set of all valid input values (x -values).
- The **range** is the set of all possible output values ($f(x)$ -values).
- A function maps each element of the domain to **exactly one** element of the range.

Function	Natural domain	Range
$f(x) = x^2$	$x \in \mathbb{R}$	$f(x) \geq 0$
$f(x) = \sqrt{x}$	$x \geq 0$	$f(x) \geq 0$
$f(x) = \frac{1}{x}$	$x \in \mathbb{R}, x \neq 0$	$f(x) \in \mathbb{R}, f(x) \neq 0$
$f(x) = \ln x$	$x > 0$	$f(x) \in \mathbb{R}$
$f(x) = \sqrt{9 - x^2}$	$-3 \leq x \leq 3$	$0 \leq f(x) \leq 3$

Exam Tip: Common restrictions on the domain:

- Cannot divide by zero: exclude values making the denominator zero.

- Cannot take the square root of a negative number (over the reals): set the expression ≥ 0 .
- Cannot take $\ln$ of a non-positive number: set the argument > 0 .

Composite Functions

The **composite function** $fg(x)$ (read " f of g of x ") means apply g first, then apply f to the result: $fg(x) = f(g(x))$. In general, $fg(x) \neq gf(x)$ — order matters.

Example: Given $f(x) = 2x + 1$ and $g(x) = x^2$, find $fg(x)$ and $gf(x)$.

$$fg(x) = f(g(x)) = f(x^2) = 2x^2 + 1$$

$$gf(x) = g(f(x)) = g(2x + 1) = (2x + 1)^2 = 4x^2 + 4x + 1$$

These are different functions.

Example: $f(x) = \sqrt{x - 1}$ (domain $x \geq 1$) and $g(x) = x^2 + 1$. Find $fg(x)$ and its domain.

$$fg(x) = f(g(x)) = f(x^2 + 1) = \sqrt{(x^2 + 1) - 1} = \sqrt{x^2} = |x|$$

The domain of fg requires $g(x) \geq 1$, i.e. $x^2 + 1 \geq 1$, which is true for all $x \in \mathbb{R}$.

So $fg(x) = |x|$, domain $x \in \mathbb{R}$, range $fg(x) \geq 0$.

Exam Tip: The domain of fg is restricted by two conditions: (1) x must be in the domain of g , and (2) $g(x)$ must be in the domain of f . Apply both conditions and take the intersection.

Inverse Functions

The **inverse function** $f^{-1}(x)$ reverses the action of f . It exists only when f is a **one-to-one** (injective) function — every output value corresponds to exactly one input value.

Key properties:

- $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$
- The domain of f^{-1} is the range of f , and vice versa.
- The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$.

Finding an Inverse Function

1. Write $y = f(x)$.
2. Rearrange to make x the subject.
3. Replace x with $f^{-1}(x)$ and y with x .
4. State the domain of f^{-1} (= range of f).

Example: Find $f^{-1}(x)$ for $f(x) = 3x - 2$.

Let $y = 3x - 2$. Rearrange: $x = \frac{y + 2}{3}$.

So $f^{-1}(x) = \frac{x + 2}{3}$, domain $x \in \mathbb{R}$.

Check: $f(f^{-1}(x)) = 3 \cdot \frac{x + 2}{3} - 2 = x + 2 - 2 = x \checkmark$

Example: $f(x) = \frac{2x + 1}{x - 3}$ ($x \neq 3$). Find $f^{-1}(x)$.

Let $y = \frac{2x + 1}{x - 3}$. Cross multiply: $y(x - 3) = 2x + 1$.

$xy - 3y = 2x + 1 \Rightarrow xy - 2x = 3y + 1 \Rightarrow x(y - 2) = 3y + 1$.

$x = \frac{3y + 1}{y - 2}$

So $f^{-1}(x) = \frac{3x+1}{x-2}$, domain $x \neq 2$.

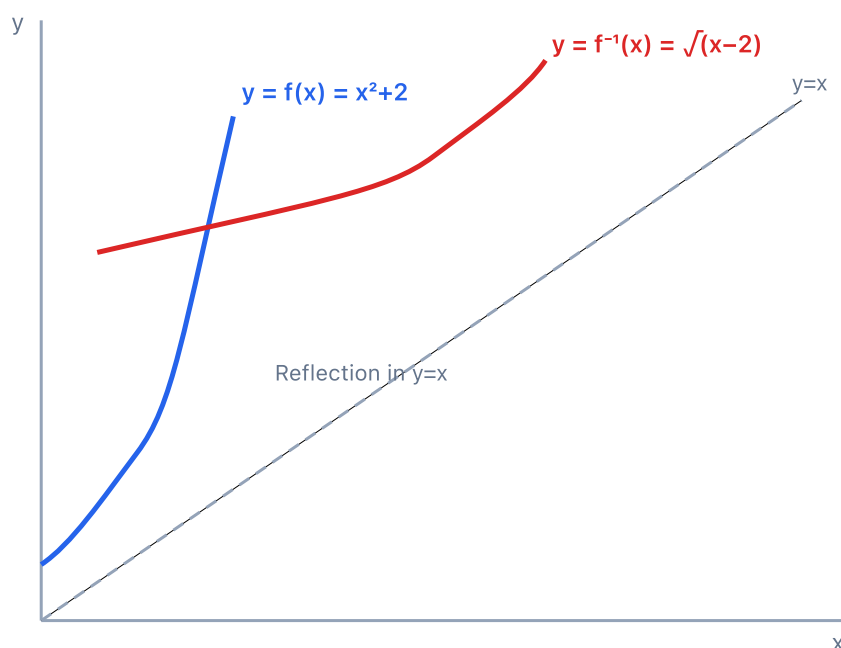
Note: f^{-1} has the same algebraic form structure as f (a common feature of this type of function).

Example: $f(x) = x^2 + 2$, domain $x \geq 0$. Find $f^{-1}(x)$ and its domain.

The domain restriction $x \geq 0$ makes f one-to-one (without it, f would not have an inverse).

Let $y = x^2 + 2$. Then $x = \sqrt{y-2}$ (taking positive root since $x \geq 0$).

$f^{-1}(x) = \sqrt{x-2}$, domain: $x \geq 2$ (= range of f).



Exam Tip: A function only has an inverse if it is **one-to-one**. For many-to-one functions (like x^2 on all reals), you must restrict the domain first. The examiner may give you a domain — use it. Always state the domain of the inverse function.

Self-Inverse Functions

A function is **self-inverse** if $f^{-1}(x) = f(x)$, i.e. $f(f(x)) = x$. Examples include $f(x) = \frac{1}{x}$ and $f(x) = \frac{a}{x}$. Their graphs are symmetric about $y = x$.